```
    separate ( IDL.SEM_PHASE )¶    --|----------------------------------------------------------------------------------------------¶    --|¬SEM_GLOB¶
    --|----------------------------------------------------------------------------------------------¶    package body SEM_GLOB is¶     use¬EXPRESO;¶
    ¶      --|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||¶      --|¶       procedure INITIALIZE_GLOBAL
_DATA¬is¶      begin¶      ¶         SU.USED_PACKAGE_LIST¬¬:= (TREE_NIL,TREE_NIL);¶         SU.INCOMPLETE_TYPE_LIST¬:= (TREE_NIL,TREE_NIL);¶         S
U.PRIVATE_TYPE_LIST¬¬:= (TREE_NIL,TREE_NIL);¶      ¶         INITIAL_H := (¬REGION_DEF¬=> TREE_VOID,¶¬¬¬LEX_LEVEL¬¬=> 1,¶¬¬¬IS_IN_SPEC¬=> TRUE,¶¬¬¬IS_
IN_BODY¬=> FALSE,¶¬¬¬SUBP_SYMREP¬=> TREE_VOID,¶¬¬¬RETURN_TYPE¬=> TREE_VOID,¶¬¬¬ENCLOSING_LOOP_ID¬=> TREE_VOID¶¬¬);¶      end¬INITIALIZE_GLOBAL_DATA;¶
      --||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||¶      --|¶      procedure INITIALIZE_PREDEFINED_T
YPES is¶         --|----------------------------------------------------------------------------------------¶         --|¬PROCEDURE¬INIT¶¬function INI
T ( X :STRING ) return TREE¬is¶¬  DEFLIST¬¬: SEQ_TYPE¬:= LIST (¬STORE_SYM¬( X ) );¶¬  DEF¬¬: TREE;¶         begin¶¬  while not IS_EMPTY ( DEFLIST ) lo
op¶¬     POP ( DEFLIST,¬DEF );¶¬     if D¬( XD_REGION_DEF, DEF ) = PREDEFINED_STANDARD_DEF then¶¬        return D ( SM_TYPE_SPEC, D ( XD_SOURCE_NAME,
DEF ) );¶¬     end if;¶¬  end loop;¶¬  return TREE_VOID;¶         end INIT;¶         --|----------------------------------------------------------------------------
--------------------------¶         --|¬PROCEDURE¬SET_RANGE_BOUNDS¶¬procedure¬SET_RANGE_BOUNDS ( TYPE_NODE :TREE; LOW, HIGH :out TREE ) is¶¬  RANGE_NO
DE: TREE;¶         begin¶¬  if TYPE_NODE /= TREE_VOID then¶¬     RANGE_NODE := D ( SM_RANGE, TYPE_NODE );¶¬     LOW¬:= EXPRESO.GET_STATIC_VALUE (¬D (
AS_EXP1, RANGE_NODE ) );¶¬     HIGH¬:= EXPRESO.GET_STATIC_VALUE (¬D ( AS_EXP2, RANGE_NODE ) );¶¬  else¶¬     LOW¬:= UARITH.U_VAL ( 0¬);¶¬     HIGH¬:=
UARITH.U_VAL ( 0¬);¶¬  end if;¶         end SET_RANGE_BOUNDS;¶         --|----------------------------------------------------------------------------
------------¶         --|¬PROCEDURE¬SET_ACCURACY¶¬procedure¬SET_ACCURACY ( TYPE_NODE :TREE; ACCURACY :out TREE ) is¶         begin¶¬  if TYPE_NODE /=
TREE_VOID then¶¬     ACCURACY := D ( SM_ACCURACY, TYPE_NODE );¶¬  else¶¬     ACCURACY := UARITH.U_VAL¬( 0 );¶¬  end if;¶         end SET_ACCURACY;¶
   ¶      begin¶         PREDEFINED_BOOLEAN¬¬:= INIT (¬"BOOLEAN"¬);¶         PREDEFINED_SHORT_INTEGER¬:= INIT (¬"SHORT_INTEGER" );¶         PREDEFINED
_INTEGER¬¬:= INIT (¬"INTEGER"¬);¶         PREDEFINED_LONG_INTEGER¬:= INIT (¬"LONG_INTEGER" );¶         PREDEFINED_FLOAT¬¬:= INIT (¬"FLOAT" );¶
 PREDEFINED_LONG_FLOAT¬:= INIT (¬"LONG_FLOAT" );¶         PREDEFINED_STRING¬¬:= INIT (¬"STRING" );¶         PREDEFINED_DURATION¬¬:= INIT (¬"DURATION"
);¶         PREDEFINED_ADDRESS¬¬:= INIT (¬"_ADDRESS" );¶      ¶         PREDEFINED_STANDARD_ID := D ( XD_SOURCE_NAME, PREDEFINED_STANDARD_DEF );¶
 ¶¶¶¶         SET_RANGE_BOUNDS ( PREDEFINED_SHORT_INTEGER, PREDEFINED_SHORT_INTEGER_FIRST, PREDEFINED_SHORT_INTEGER_LAST );¶         SET_RANGE_BOUNDS
( PREDEFINED_INTEGER, PREDEFINED_INTEGER_FIRST, PREDEFINED_INTEGER_LAST );¶         SET_RANGE_BOUNDS ( PREDEFINED_LONG_INTEGER, PREDEFINED_LONG_INTEGE
R_FIRST, PREDEFINED_LONG_INTEGER_LAST );¶         if PREDEFINED_LONG_INTEGER /= TREE_VOID then¶¬  PREDEFINED_LARGEST_INTEGER := PREDEFINED_LONG_INTEGE
R;¶         else¶¬  PREDEFINED_LARGEST_INTEGER := PREDEFINED_INTEGER;¶         end if;¶      ¶         SET_RANGE_BOUNDS ( PREDEFINED_FLOAT, PREDEFINED
_FLOAT_FIRST,¬PREDEFINED_FLOAT_LAST );¶         SET_ACCURACY ( PREDEFINED_FLOAT, PREDEFINED_FLOAT_ACCURACY);¶         SET_RANGE_BOUNDS ( PREDEFINED_LO
NG_FLOAT, PREDEFINED_LONG_FLOAT_FIRST,¬PREDEFINED_LONG_FLOAT_LAST );¶         SET_ACCURACY ( PREDEFINED_LONG_FLOAT, PREDEFINED_LONG_FLOAT_ACCURACY );¶
         if PREDEFINED_LONG_FLOAT /= TREE_VOID then¶¬  PREDEFINED_LARGEST_FLOAT :=¬PREDEFINED_LONG_FLOAT;¶         else¶¬  PREDEFINED_LARGEST_FLOAT :=
¬PREDEFINED_FLOAT;¶         end if;¶      ¶      end¬INITIALIZE_PREDEFINED_TYPES;¶      ¶   --|--------------------------------------------------------
---------------------------------------¶   end SEM_GLOB;¶
```